

PAPER_1660_HUBBARD_U_OVER_T_4

Daniel T. Murphy

PAPER_1660 — Hubbard Mott Threshold $U/t = U/t = D_{\text{phys}} = 4$ EXACT integer-primitive

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Strongly Correlated

Date: June 18, 2026

Status: CLOSED — EXACT closure (PAPER_134x broader corpus)

Observation

PAPER_1348 (PAPER_134x condensed matter / quantum sector) gives:

``

$U/t = D_{\text{phys}} = 4$ EXACT integer-primitive

``

Status: **EXACT**.

UQFF Closed Identity

``

$U/t = D_{\text{phys}} = 4$ EXACT integer-primitive EXACT

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental strongly correlated measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1348

- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)

- Calculator dispatch: `calculate_paradox({"paradox": "hubbard_u_over_t_4"})`

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